

Structural Stability: Persistence of Constraint Satisfaction in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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April 15, 2026 — Version 1

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Abstract

This document defines structural stability as the persistence of the relation between load and capacity under boundary conditions over time in Dimensional Human Field Theory (DHFT).

Structural stability is maintained when load remains processable relative to capacity under boundary conditions across an interval, not solely at a single measurement point.

I. Scope

This document defines structural stability as a condition of system behavior across time.

It distinguishes stability as persistence of constraint satisfaction, not an instantaneous state.

II. Definition

Structural stability is defined as the persistence of the relation:

$$L \leq C$$

under boundary conditions across an interval of time.

III. Temporal Condition

Structural stability requires that constraint satisfaction is maintained over time.

Structural instability results from constraint violation across an interval, even if instantaneous measurements appear stable.

IV. Non-Instantaneous Condition

Stability is not determined by a single measurement point.

Apparent stability at a moment coincides with accumulated load sufficient to produce instability.

V. Relation to Identity

Structural stability is a condition of system state.

Identity is defined by structural continuity and is independent of stability.

Loss of stability occurs while identity is retained.

VI. Relation to Drift and Recovery

- Structural Drift is deviation from constraint conditions while structural continuity remains
- Structural Recovery is restoration of capacity and boundary stabilization under load

Structural stability is maintained when constraint conditions remain satisfied without deviation.

VII. Constraint

Structural stability is defined over the invariant variables:

- Load (L)
- Capacity (C)
- Boundary (B)

No additional variables are introduced.

VIII. Statement

Structural stability is the persistence of constraint satisfaction over time under boundary conditions.

IX. System Reference

The system defined in this document is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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